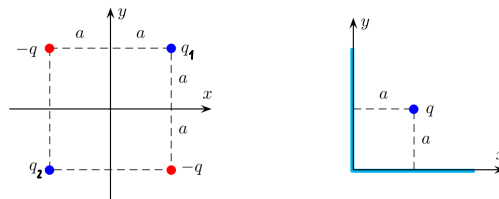


Opgave 1

Fire punktladninger er anbragt i xy -planet af et koordinatsystem som vist på figuren til venstre. To ladninger, q , er anbragt i punkterne $(a, a, 0)$ hhv. $(-a, -a, 0)$, mens to ladninger, $-q$, er anbragt i punkterne $(a, -a, 0)$ hhv. $(-a, a, 0)$.



a) Bestem kraften, $\mathbf{F}$, på ladningen q i punktet $(a, a, 0)$. Resultatet bedes givet på vektorform.

Der betragtes derefter situationen illustreret på figuren til højre. En punktladning, q , er anbragt i punktet $(a, a, 0)$ i det vinkelrette hjørne dannet af to ledende planer af uendelig udstrækning. De to planer ligger henholdsvis i planerne udtrakt af xz - og yz -akserne, og står altså vinkelret på figurens plan. Systemet af de to planer har potentialet nul.

$$\hat{r}_{12} = 2a\hat{x} + 2a\hat{y}, \quad r = \sqrt{(2a)^2 + (2a)^2} = \sqrt{8}a = 2\sqrt{2}a, \quad \hat{r} = \frac{\hat{x} + \hat{y}}{\sqrt{2}}$$

$$\vec{F} = \frac{q^2}{4\pi\epsilon_0} \left(\frac{-1}{(2a)^2} (\hat{x} + \hat{y}) + \frac{1}{(\sqrt{8}a)^2} \frac{\hat{x} + \hat{y}}{\sqrt{2}} \right) = \frac{1}{4\pi\epsilon_0} \frac{q^2}{a^2} \left(\frac{1}{\sqrt{2}8} - \frac{1}{4} \right) (\hat{x} + \hat{y}) = \frac{1}{4\pi\epsilon_0} \frac{q^2}{16a^2} (\sqrt{2} - 4) (\hat{x} + \hat{y})$$

b) Argumentér for, at resultatet fundet under a) også er gyldigt for denne situation.

b) På figuren til venstre har planerne $x = 0$ og $y = 0$ elektrisk potential lig nul. Dette er det samme som på figuren til højre (for $x \geq 0$ og $y \geq 0$). I første kvadrant har potentialet dermed identiske grænsebetingelser i de to situationer. Ifølge entydighedsteoremet er potentialet, og dermed feltet og kraften på q det samme i de to situationer.

Opgave 5

En elektrostatisk opstilling består af en tynd kugleskal med radius R . Det elektriske potential på kugleskallen er

$$\begin{aligned} V(\theta) &= 2V_0 P_2(\cos\theta) \\ &= V_0 (3\cos^2\theta - 1), \end{aligned}$$

hvor V_0 er en konstant, og $P_2(\cos\theta) = (3\cos^2\theta - 1)/2$ er det andet Legendre-polynomium. Potentialet er nul i det uendeligt fjerne.

a) Argumentér for, at det elektriske potential overalt uden for kugleskallen ($r > R$) er

$$V(r, \theta) = V_0 (3\cos^2\theta - 1) \left(\frac{R}{r} \right)^3.$$

Idet potentiallet på kugleskallen kun bruger leddet for $\ell=2$ reduceres det generelle udtryk $V(r, \theta) = \sum_{\ell=0}^{\infty} \left(A_{\ell} r^{\ell} + \frac{B_{\ell}}{r^{\ell+1}} \right) P_{\ell}(\cos\theta)$ til $V(r, \theta) = \left(A_2 r^2 + \frac{B_2}{r^3} \right) P_2(\cos\theta)$. For $r > R$ vil det første led $A_2 r^2$ divergere for $r \rightarrow \infty$, hvorfor $A_2 = 0$. Idet $V(r, \theta) = 2V_0 P_2(\cos\theta)$, når $r = R$ må $B_2 = 2V_0 R^3$ og $V(r, \theta) = 2V_0 \frac{R^3}{r^3} P_2(\cos\theta) = V_0 (3\cos^2\theta - 1) \left(\frac{R}{r} \right)^3$ for $r > R$.

b) Bestem det elektriske potential overalt inden for kugleskallen ($r < R$).

Her vil $\frac{B_2}{r^3}$ divergere for $r \rightarrow 0$, hvorfor $B_2 = 0$. Igen må $V(r = R, \theta) = 2V_0 P_2(\cos\theta)$, så $A_2 = \frac{2V_0}{R^2}$ og $V(r, \theta) = V_0 (3\cos^2\theta - 1) \left(\frac{r}{R} \right)^2$ for $r < R$.

c) Bestem det elektriske felt overalt uden for kugleskallen ($r > R$). Angiv resultatet på vektorform i polære koordinater, $\mathbf{E}(r, \theta) = E_r \hat{r} + E_{\theta} \hat{\theta} + E_{\phi} \hat{\phi}$.

$$\vec{E} = -\vec{\nabla}V = -\frac{\partial V}{\partial r} \hat{r} - \frac{1}{r} \frac{\partial V}{\partial \theta} \hat{\theta} = 3V_0 (3\cos^2\theta - 1) \frac{R^3}{r^4} \hat{r} + V_0 (6\cos\theta \sin\theta) \frac{R^3}{r^4} \hat{\theta} = \frac{3V_0 R^3}{r^4} [(3\cos^2\theta - 1) \hat{r} + 2\cos\theta \sin\theta \hat{\theta}]$$

Problem 3.18 The potential at the surface of a sphere (radius R) is given by

$$V_0 = k \cos 3\theta,$$

where k is a constant. Find the potential inside and outside the sphere, as well as the surface charge density $\sigma(\theta)$ on the sphere. (Assume there's no charge inside or outside the sphere.)

trig. identitet

$$V_0(\theta) = k \cos 3\theta \stackrel{\text{trig. identitet}}{=} k(4 \cos^3 \theta - 3 \cos \theta) = k \left(\frac{8}{5} P_3(\cos \theta) - \frac{3}{5} P_1(\cos \theta) \right)$$

Nu har vi
$$V(r, \theta) = \begin{cases} \sum_{\ell=0}^{\infty} A_{\ell} r^{\ell} P_{\ell}(\cos \theta), & \text{for } r < R \quad (3.66) \\ \sum_{\ell=0}^{\infty} \frac{B_{\ell}}{r^{\ell+1}} P_{\ell}(\cos \theta), & \text{for } r > R \quad (3.72) \end{cases}$$

(3.69) giver
$$A_{\ell} = \frac{2\ell+1}{2R^{\ell}} \int_0^{\pi} V_0(\theta) P_{\ell}(\cos \theta) \sin \theta d\theta = \frac{2\ell+1}{2R^{\ell}} \frac{k}{5} \left(\int_0^{\pi} 8 P_3(\cos \theta) P_{\ell}(\cos \theta) \sin \theta d\theta - \int_0^{\pi} 3 P_1(\cos \theta) P_{\ell}(\cos \theta) \sin \theta d\theta \right)$$

(3.68)
$$\stackrel{\text{trig. identitet}}{=} \frac{2\ell+1}{2R^{\ell}} \frac{k}{5} \left(8 \frac{2}{2\ell+1} \delta_{\ell 3} - 3 \frac{2}{2\ell+1} \delta_{\ell 1} \right) = \frac{k}{5R^{\ell}} (8\delta_{\ell 3} - 3\delta_{\ell 1}) =$$

$$\begin{cases} \frac{8k}{5R^3}, & \text{hvis } \ell=3 \\ \frac{-3k}{5R}, & \text{hvis } \ell=1 \end{cases}$$
 for alle andre værdier er $A_{\ell}=0$

$$V(r, \theta) = A_1 r^1 P_1(\cos \theta) + A_3 r^3 P_3(\cos \theta) = \frac{8}{5} k \left(\frac{r}{R} \right)^3 P_3(\cos \theta) - \frac{3}{5} k \frac{r}{R} P_1(\cos \theta), \text{ for } r < R$$

(3.81)
$$B_{\ell} = A_{\ell} R^{2\ell+1}. \quad B_1 = A_1 R^3 = \frac{-3k}{5R} R^3 = \frac{-3}{5} k R^2, \quad B_3 = A_3 R^7 = \frac{8}{5} k R^4$$

$$V(r, \theta) = \frac{B_1}{r^2} P_1(\cos \theta) + \frac{B_3}{r^4} P_3(\cos \theta) = \frac{8}{5} k \left(\frac{R}{r} \right)^4 P_3(\cos \theta) - \frac{3}{5} k \left(\frac{R}{r} \right)^2 P_1(\cos \theta), \text{ for } r > R$$

(3.83)
$$\sigma_0(\theta) = \epsilon_0 \sum_{\ell=0}^{\infty} (2\ell+1) A_{\ell} R^{\ell-1} P_{\ell}(\cos \theta) = \epsilon_0 (3A_1 P_1(\cos \theta) + 7A_3 R^2 P_3(\cos \theta)) = \epsilon_0 \left(\frac{-3k}{5R} 3P_1(\cos \theta) + \right.$$

$$\left. 7 \frac{8k}{5R^3} R^2 P_3(\cos \theta) \right) = \frac{k\epsilon_0}{5R} (56 P_3(\cos \theta) - 9 P_1(\cos \theta))$$

Problem 3.28 In Ex. 3.9 we derived the exact potential for a spherical shell of radius R , which carries a surface charge $\sigma = k \cos \theta$.

(a) Calculate the dipole moment of this charge distribution.

(b) Find the approximate potential, at points far from the sphere, and compare the exact answer (3.87). What can you conclude about the higher multipoles?

a) Grundet Symmetri må $\vec{P} = P\hat{z}$. $P = \int z \rho d\tau = \int z \sigma da$, $z = r \cos \theta$, $da = r^2 \sin \theta d\theta d\phi$

$$P = \int_0^{2\pi} \int_0^\pi R \cos \theta k \cos \theta R^2 \sin \theta d\theta d\phi = k R^3 2\pi \int_0^\pi \cos^2 \theta \sin \theta d\theta = 2\pi k R^3 \left[-\frac{\cos^3 \theta}{3} \right]_0^\pi = 2\pi k R^3 \left[\frac{1}{3} + \frac{1}{3} \right] = \frac{4\pi k R^3}{3}$$

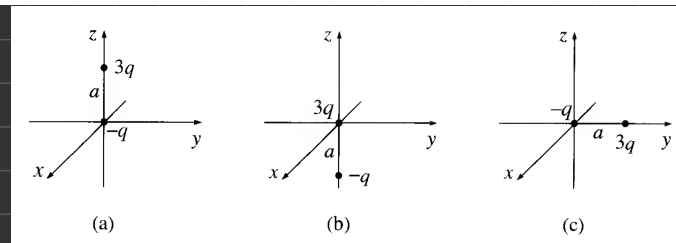
$$\vec{P} = \frac{4\pi k R^3}{3} \hat{z}$$

b)

$$V \approx \frac{1}{4\pi\epsilon_0} \frac{\vec{P} \cdot \hat{r}}{r^2} = \frac{1}{4\pi\epsilon_0} \frac{4\pi k R^3}{3} \frac{\cos \theta}{r^2} = \frac{k R^3 \cos \theta}{3\epsilon_0 r^2}$$

Dette er det eksakte potential, så alle andre multipol momenter end dipolmomentet må være nul

Problem 3.30 Two point charges, $3q$ and $-q$, are separated by a distance a . For each of the arrangements in Fig. 3.35, find (i) the monopole moment, (ii) the dipole moment, and (iii) the approximate potential (in spherical coordinates) at large r (include both the monopole and dipole contributions).



a) $Q = 2q$, $\vec{P} = 0 + 3qa\hat{z} = 3qa\hat{z}$, $V \approx \frac{1}{4\pi\epsilon_0} \left[\frac{Q}{r} + \frac{\vec{P} \cdot \hat{r}}{r^2} \right] = \frac{1}{4\pi\epsilon_0} \left(\frac{2q}{r} + \frac{3qa \cos \theta}{r^2} \right)$

b) $Q = 2q$, $\vec{P} = -qa\hat{z}$, $V \approx \frac{1}{4\pi\epsilon_0} \left[\frac{2q}{r} - \frac{qa \cos \theta}{r^2} \right]$

c) $Q = 2q$, $\vec{P} = 3qa\hat{y}$, $V \approx \frac{1}{4\pi\epsilon_0} \left[\frac{2q}{r} + \frac{3qa \sin \theta \sin \phi}{r^2} \right]$

Problem 3.31 A "pure" dipole p is situated at the origin, pointing in the z direction.

(a) What is the force on a point charge q at $(a, 0, 0)$ (Cartesian coordinates)?

(b) What is the force on q at $(0, 0, a)$?

(c) How much work does it take to move q from $(a, 0, 0)$ to $(0, 0, a)$?

$$\vec{E}_{\text{dip}}(r, \theta) = \frac{p}{4\pi\epsilon_0 r^3} (2 \cos \theta \hat{r} + \sin \theta \hat{\theta}), \quad \vec{F} = q\vec{E}$$

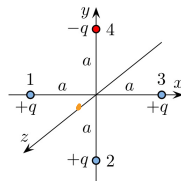
a) $r = a$, $\theta = \frac{\pi}{2}$. $\vec{F}_{\text{dip}} = \frac{qp}{4\pi\epsilon_0 a^3} (2 \cos \frac{\pi}{2} \hat{r} + \sin \frac{\pi}{2} \hat{\theta}) = \frac{qp}{4\pi\epsilon_0 a^3} \hat{\theta} = \frac{qp}{4\pi\epsilon_0 a^3} (-\hat{z})$

b) $r = a$, $\theta = 0$. $\vec{F}_{\text{dip}} = \frac{qp}{4\pi\epsilon_0 a^3} (2 \cos(0) \hat{r} + \sin(0) \hat{\theta}) = \frac{2qp}{4\pi\epsilon_0 a^3} \hat{r} = \frac{2qp}{4\pi\epsilon_0 a^3} \hat{z}$

c) $V_{\text{dip}} = \frac{p \cos \theta}{4\pi\epsilon_0 r^2}$. $W = q[V(a, 0, 0) - V(0, 0, a)] = q \frac{p}{4\pi\epsilon_0 a^2} (\cos(0) - \cos \frac{\pi}{2}) = \frac{qp}{4\pi\epsilon_0 a^2}$

Opgave 1

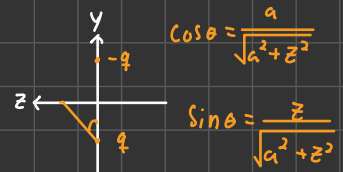
Fire punktladninger er placeret i xy -planet i et kartesisk koordinatsystem, alle i afstanden a fra origo: Tre identiske punktladninger, $+q$, er placeret i punkterne $(-a, 0, 0)$, $(0, -a, 0)$ og $(a, 0, 0)$; en fjerde punktladning med modsat ladning, $-q$, er placeret i punktet $(0, a, 0)$.



a) Bestem den elektrostatiske energi af ladningssystemet.

$$(2.40) \quad W = \frac{q^2}{4\pi\epsilon_0} \left(\frac{1}{r_{12}} + \frac{1}{r_{13}} - \frac{1}{r_{14}} + \frac{1}{r_{23}} - \frac{1}{r_{24}} - \frac{1}{r_{34}} \right) = \frac{1}{4\pi\epsilon_0} \frac{q^2}{a} \left(\frac{1}{\sqrt{2}} + \frac{1}{2} - \frac{1}{\sqrt{2}} + \frac{1}{\sqrt{2}} - \frac{1}{2} - \frac{1}{\sqrt{2}} \right) = 0$$

b) Bestem det elektriske felt først i origo og dernæst i et vilkårligt punkt $(0, 0, z)$ på z -aksen. Resultatet bedes i begge tilfælde angivet på vektorform, $\mathbf{E}(z) = E_x(z) \hat{\mathbf{x}} + E_y(z) \hat{\mathbf{y}} + E_z(z) \hat{\mathbf{z}}$.



$$\vec{E}(z=0) = \frac{1}{4\pi\epsilon_0} \left(\frac{q}{a^2} \hat{\mathbf{y}} + \frac{-q}{a^2} (-\hat{\mathbf{y}}) \right) = \frac{1}{4\pi\epsilon_0} \frac{2q}{a^2} \hat{\mathbf{y}}$$

$$\vec{E}(z) = \frac{1}{4\pi\epsilon_0} \left(\frac{2q}{\sqrt{a^2 + z^2}} \cos\theta \hat{\mathbf{y}} + \frac{2q}{\sqrt{a^2 + z^2}} \sin\theta \hat{\mathbf{z}} \right) = \frac{1}{4\pi\epsilon_0} \frac{2q}{(a^2 + z^2)^{3/2}} (a \hat{\mathbf{y}} + z \hat{\mathbf{z}})$$

c) Bestem ladningssystemets dipolmoment $\mathbf{p}$. Opskriv dernæst et tilnærmet udtryk for det elektriske potential, som er gyldigt i ethvert punkt $\mathbf{r}$ langt væk fra origo ($r = |\mathbf{r}| \gg a$). Udtrykket skal indholde bidrag af orden op til, og inklusive, dipolledet. Potentialet sættes til nul i det uendeligt fjerne. Angiv resultatet i sfæriske koordinater.

$$\vec{p} = \sum_{i=1}^4 q_i \vec{r}_i' = qa \hat{\mathbf{y}} + qa(-\hat{\mathbf{y}}) - qa \hat{\mathbf{y}} + qa(-\hat{\mathbf{y}}) = -2qa \hat{\mathbf{y}}, \quad Q = 2q$$

$$V_{\text{mon}}(\vec{r}) = \frac{1}{4\pi\epsilon_0} \frac{2q}{r} \quad V_{\text{dip}}(\vec{r}) = \frac{1}{4\pi\epsilon_0} \frac{\vec{p} \cdot \vec{r}}{r^2} = \frac{-1}{4\pi\epsilon_0} \frac{2qa}{r^2} \hat{\mathbf{y}} \cdot \vec{r} = \frac{-1}{4\pi\epsilon_0} \frac{2qa}{r^2} \sin\theta \sin\phi$$

$$V(\vec{r}) = V_{\text{mon}}(\vec{r}) + V_{\text{dip}}(\vec{r}) = \frac{2q}{4\pi\epsilon_0} \left(\frac{1}{r} - \frac{2a}{r^2} \sin\theta \sin\phi \right)$$

Problem 3.33 Show that the electric field of a ("pure") dipole (Eq. 3.103) can be written in the coordinate-free form

$$\mathbf{E}_{\text{dip}}(\mathbf{r}) = \frac{1}{4\pi\epsilon_0} \frac{1}{r^3} [3(\mathbf{p} \cdot \hat{\mathbf{r}})\hat{\mathbf{r}} - \mathbf{p}]. \quad (3.104)$$

(3.103) $\bar{\mathbf{E}}_{\text{dip}}(r, \theta) = \frac{p}{4\pi\epsilon_0 r^3} (2\cos\theta \hat{\mathbf{r}} + \sin\theta \hat{\boldsymbol{\theta}})$. Vi lagger koordinatsystemet så $\bar{\mathbf{p}} = p\hat{\mathbf{z}}$

$\hat{\mathbf{z}} = \cos\theta \hat{\mathbf{r}} - \sin\theta \hat{\boldsymbol{\theta}}$. $\bar{\mathbf{p}} \cdot \hat{\mathbf{r}} = p\hat{\mathbf{z}} \cdot \hat{\mathbf{r}} = p\cos\theta$, $3(\bar{\mathbf{p}} \cdot \hat{\mathbf{r}})\hat{\mathbf{r}} = 3p\cos\theta \hat{\mathbf{r}}$, $\bar{\mathbf{p}} = p(\cos\theta \hat{\mathbf{r}} - \sin\theta \hat{\boldsymbol{\theta}})$

$\frac{1}{4\pi\epsilon_0 r^3} (3(\bar{\mathbf{p}} \cdot \hat{\mathbf{r}})\hat{\mathbf{r}} - \bar{\mathbf{p}}) = \frac{1}{4\pi\epsilon_0 r^3} (3p\cos\theta \hat{\mathbf{r}} - p\cos\theta \hat{\mathbf{r}} + p\sin\theta \hat{\boldsymbol{\theta}}) = \frac{p}{4\pi\epsilon_0 r^3} (2\cos\theta \hat{\mathbf{r}} + \sin\theta \hat{\boldsymbol{\theta}})$ ✓

Problem 4.5 In Fig. 4.6, $\mathbf{p}_1$ and $\mathbf{p}_2$ are (perfect) dipoles a distance r apart. What is the torque on $\mathbf{p}_1$ due to $\mathbf{p}_2$? What is the torque on $\mathbf{p}_2$ due to $\mathbf{p}_1$? [In each case I want the torque on the dipole about its own center. If it bothers you that the answers are not equal and opposite, see Prob. 4.29.]

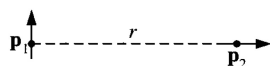


Figure 4.6

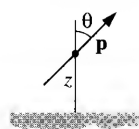


Figure 4.7

$\bar{\mathbf{p}}_1 = p_1 \hat{\mathbf{z}}$, $\bar{\mathbf{p}}_2 = p_2 \hat{\mathbf{x}}$, fra $\bar{\mathbf{p}}_1$ til $\bar{\mathbf{p}}_2$ er $\hat{\mathbf{r}}_2 = \hat{\mathbf{x}}$ og fra $\bar{\mathbf{p}}_2$ til $\bar{\mathbf{p}}_1$ er $\hat{\mathbf{r}}_1 = -\hat{\mathbf{x}}$

$\bar{\mathbf{E}}_1 = \frac{1}{4\pi\epsilon_0 r^3} (3(\bar{\mathbf{p}}_2 \cdot \hat{\mathbf{r}}_1)\hat{\mathbf{r}}_1 - \bar{\mathbf{p}}_2) = \frac{1}{4\pi\epsilon_0 r^3} (3(p_2 \hat{\mathbf{x}} \cdot (-\hat{\mathbf{x}}))(-\hat{\mathbf{x}}) - p_2 \hat{\mathbf{x}}) = \frac{-p_2 \hat{\mathbf{x}}}{4\pi\epsilon_0 r^3}$

(4.4) $\bar{\mathbf{N}}_2 = \bar{\mathbf{p}}_2 \times \bar{\mathbf{E}}_1 = (p_2 \hat{\mathbf{x}}) \times \left(\frac{-p_1 \hat{\mathbf{z}}}{4\pi\epsilon_0 r^3} \right) = \frac{-p_1 p_2}{4\pi\epsilon_0 r^3} (\hat{\mathbf{x}} \times \hat{\mathbf{z}}) = \frac{-p_1 p_2}{4\pi\epsilon_0 r^3} (-\hat{\mathbf{y}}) = \frac{p_1 p_2}{4\pi\epsilon_0 r^3} \hat{\mathbf{y}}$

$\bar{\mathbf{E}}_2 = \frac{1}{4\pi\epsilon_0 r^3} (3(\bar{\mathbf{p}}_1 \cdot \hat{\mathbf{r}}_2)\hat{\mathbf{r}}_2 - \bar{\mathbf{p}}_1) = \frac{1}{4\pi\epsilon_0 r^3} (3(p_1 \hat{\mathbf{z}} \cdot \hat{\mathbf{x}})\hat{\mathbf{x}} - p_1 \hat{\mathbf{z}}) = \frac{1}{4\pi\epsilon_0 r^3} (2p_1 \hat{\mathbf{x}} - p_1 \hat{\mathbf{z}})$

$\bar{\mathbf{N}}_1 = \bar{\mathbf{p}}_1 \times \bar{\mathbf{E}}_2 = (p_1 \hat{\mathbf{z}}) \times \left(\frac{2p_2 \hat{\mathbf{x}}}{4\pi\epsilon_0 r^3} \right) = \frac{2p_1 p_2}{4\pi\epsilon_0 r^3} (\hat{\mathbf{z}} \times \hat{\mathbf{x}}) = \frac{2p_1 p_2}{4\pi\epsilon_0 r^3} \hat{\mathbf{y}}$

Problem 4.8 Show that the interaction energy of two dipoles separated by a displacement $\mathbf{r}$ is

$$U = \frac{1}{4\pi\epsilon_0} \frac{1}{r^3} [\mathbf{p}_1 \cdot \mathbf{p}_2 - 3(\mathbf{p}_1 \cdot \hat{\mathbf{r}})(\mathbf{p}_2 \cdot \hat{\mathbf{r}})]. \quad (4.7)$$

[Hint: use Prob. 4.7 and Eq. 3.104.]

$\bar{\mathbf{c}}(\bar{\mathbf{a}} \cdot \bar{\mathbf{b}})\bar{\mathbf{d}} = (\bar{\mathbf{c}} \cdot \bar{\mathbf{b}}) \cdot (\bar{\mathbf{c}} \cdot \bar{\mathbf{d}})$

(4.6) $U = -\bar{\mathbf{p}}_1 \cdot \bar{\mathbf{E}}_2 = -\bar{\mathbf{p}}_1 \cdot \frac{1}{4\pi\epsilon_0 r^3} (3(\bar{\mathbf{p}}_2 \cdot \hat{\mathbf{r}})\hat{\mathbf{r}} - \bar{\mathbf{p}}_2) = \frac{1}{4\pi\epsilon_0 r^3} (\bar{\mathbf{p}}_1 \cdot \bar{\mathbf{p}}_2 - 3(\bar{\mathbf{p}}_1 \cdot \hat{\mathbf{r}})(\bar{\mathbf{p}}_2 \cdot \hat{\mathbf{r}}))$

Problem 4.10 A sphere of radius R carries a polarization

$$\mathbf{P}(\mathbf{r}) = k\mathbf{r},$$

where k is a constant and $\mathbf{r}$ is the vector from the center.

(a) Calculate the bound charges σ_b and ρ_b .

(b) Find the field inside and outside the sphere.

$$\vec{P} = k \vec{r} = k r \hat{r}$$

$$a) \sigma_b = \vec{P} \cdot \hat{n} = k \vec{r} \cdot \hat{r} = k R \hat{r} \cdot \hat{r} = k R, \quad \rho_b = -\nabla \cdot \vec{P} = -\frac{1}{r^2} \frac{\partial}{\partial r} (r^2 k r) = -\frac{k}{r^2} 3r^2 = -3k$$

$$b) (2.13) \oint \vec{E} \cdot d\vec{a} = \frac{Q_{enc}}{\epsilon_0}$$

$$r < R \quad Q_{enc} = \frac{4}{3} \pi r^3 \rho_b = \frac{4}{3} \pi r^3 (-3k) = -4\pi k r^3. \quad \oint \vec{E} \cdot d\vec{a} = E 4\pi r^2 = \frac{-4\pi k r^3}{\epsilon_0} \Rightarrow \vec{E} = -\frac{k r}{\epsilon_0} \hat{r}$$

$$r > R \quad Q_{enc} = 4\pi R^2 \sigma_b + \frac{4}{3} \pi R^3 \rho_b = 4\pi R^2 k R - 4\pi k R^3 = 0 \Rightarrow \vec{E} = \vec{0}$$

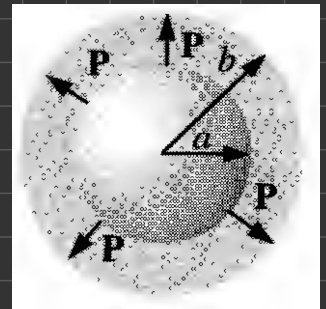
Problem 4.15 A thick spherical shell (inner radius a , outer radius b) is made of dielectric material with a "frozen-in" polarization

$$\mathbf{P}(\mathbf{r}) = \frac{k}{r} \hat{r},$$

where k is a constant and r is the distance from the center (Fig. 4.18). (There is no free charge in the problem.) Find the electric field in all three regions by two different methods:

(a) Locate all the bound charge, and use Gauss's law (Eq. 2.13) to calculate the field it produces.

(b) Use Eq. 4.23 to find $\mathbf{D}$, and then get $\mathbf{E}$ from Eq. 4.21. [Notice that the second method is much faster, and avoids any explicit reference to the bound charges.]



$$a) \rho_b = -\nabla \cdot \vec{P} = -\frac{1}{r^2} \frac{\partial}{\partial r} (r^2 \frac{k}{r}) = -\frac{k}{r^2}, \quad \sigma_{b_a} = \vec{P} \cdot (-\hat{r}) = -\frac{k}{a}, \quad \sigma_{b_b} = \vec{P} \cdot (\hat{r}) = \frac{k}{b}$$

$$\text{Gauss' law} \Rightarrow \vec{E} = \frac{1}{4\pi\epsilon_0} \frac{Q_{enc}}{r^2}, \quad \text{For } r < a \quad Q_{enc} = 0 \Rightarrow \vec{E} = \vec{0}$$

$$\text{For } a < r < b \quad Q_{enc} = 4\pi a^2 \left(-\frac{k}{a}\right) + \int_a^r \frac{-k}{r'^2} 4\pi r'^2 dr' = -4\pi k a - 4\pi k \int_a^r dr' = -4\pi k a - 4\pi k (r - a) = -4\pi k r$$

$$\Rightarrow \vec{E} = \frac{1}{4\pi\epsilon_0} \frac{-4\pi k r}{r^2} \hat{r} = -\frac{k}{\epsilon_0 r} \hat{r}$$

$$\text{For } r > b \quad Q_{enc} = -4\pi k b + 4\pi b^2 \frac{k}{b} = 0 \Rightarrow \vec{E} = \vec{0}$$

$$b) \oint \vec{D} \cdot d\vec{a} = Q_{fenc} = 0 \Rightarrow \vec{D} = \vec{0}, \quad \vec{D} = \epsilon_0 \vec{E} + \vec{P} = \vec{0} \Rightarrow \vec{E} = -\frac{\vec{P}}{\epsilon_0} = \begin{cases} 0 & \text{for } r < a \text{ or } r > b \\ -\frac{k}{\epsilon_0 r} \hat{r} & \text{for } a < r < b \end{cases}$$

Opgave 2

En kugle med radius R er lavet af et lineært dielektrikum med den relative permittivitet ϵ_r . Kuglen bærer en radiusafhængig volumetæthed af frie ladninger $\rho(r) = kr$, hvor k er en konstant.

a) Bestem kuglens totale frie ladning, Q .

$$Q = \int_0^R 4\pi r^2 k r dr = 4\pi \frac{R^4}{4} k = \underline{k\pi R^4}$$

b) Bestem det elektriske felt, $\mathbf{E}$, (størrelse og retning) såvel uden for ($r > R$) som inden i ($r < R$) kuglen.

Systemet er kuglesymmetrisk så $\vec{E} = \frac{1}{4\pi\epsilon} \frac{Q_{enc}}{r^2} \hat{r}$

$$\text{For } r > R \quad \underline{\vec{E} = \frac{1}{4\pi\epsilon_0} \frac{k\pi R^4}{r^2} \hat{r}}, \quad \text{for } r < R \quad \vec{E} = \frac{1}{4\pi\epsilon} \frac{Q(r)}{r^2} \hat{r} = \frac{1}{4\pi\epsilon_0\epsilon_r} \frac{k\pi r^4}{r^2} \hat{r} = \underline{\frac{1}{4\epsilon_0\epsilon_r} k r^2 \hat{r}}$$

c) Bestem dielektrikumets polarisation, $\mathbf{P}$.

$$\vec{P} = \epsilon_0 \chi_e \vec{E} = \epsilon_0 (\epsilon_r - 1) \frac{1}{4\epsilon_0\epsilon_r} k r^2 \hat{r} = \underline{\frac{\epsilon_r - 1}{\epsilon_r} \frac{k r^2}{4} \hat{r}}$$